Integrating Graph Network Annotations and Structural Similarity Data to Uncover Photosynthesis Gene Functions

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Abstract

Chlamydomonas reinhardtii, a unicellular green alga, has emerged as a valuable model organism for studying photosynthesis due to its simplicity and ease of cultivation compared to multicellular plants. Our collaborators at University of California, Berkeley, worked to identify approximately 254 high-confidence (HC) candidate photosynthesis genes from *C. reinhardtii*. To further annotate these HC genes, a multi-omics, data-driven approach has been employed, incorporating transcriptomics, proteomics, and metabolomics. This paper explores the use of *C. reinhardtii* pairwise graph networks and an Arcadia Science pipeline called ProteinCartography for structural similarity analysis. The pairwise graph networks, generated using Cytoscape and FoldSeek, cluster proteins based on structural similarity, revealing patterns and relationships that aid in identifying potential functions of HC genes. The ProteinCartography pipeline extends the structural comparison to proteins from other organisms. When combined, these methodologies help enhance our understanding of unknown photosynthesis genes, enabling the prediction of potential HC protein functions.

1 Introduction

1.1 *Chlamydomonas reinhardtii* as a Model Organism

Chlamydomonas reinhardtii is a unicellular green alga chosen as a model organism to explore the molecular components, regulation, and overall processes of photosynthesis. *C. reinhardtii's* system is much smaller, easier to cultivate, and faster to grow compared to the complex systems of multicellular plants (Sasso, 2018). This fact highlights the significant advantage in speed with which scientific studies involving *C. reinhardtii* can be conducted compared to studies using whole plants. Since this scientific realization, *C. reinhardtii* has been used as an effective model organism to advance plant and cell biology significantly (Blaby-Haas and Sabeeha, 2019). Thus, the modernization of resources used to further enable the genetic studies of *C. reinhardtii* has substantially amplified our understanding of photosynthesis on a molecular level.

1.2 Obtaining High-Confidence Photosynthesis Genes

C. reinhardtii further demonstrates its indispensable value as a tool for discovering photosynthesis genes through the sequencing of its mutants. Our collaborators from the Niyogi lab at the University of California, Berkeley, worked to sequence and analyze the genomes of 660 *C. reinhardtii* acetate-requiring mutants (Wakao et al., 2021). These mutants were generated by insertional mutagenesis using a linearized plasmid that partially or fully disrupted genes. Mutants were re-screened and re-sequenced to identify alternative carbon sources and light sensitivity. From these re-screened mutants, a set of approximately 254 high-confidence (HC) candidate photosynthesis genes were curated based on a specific criteria. This criteria includes conservation among photosynthetic organisms, possession of a domain commonly associated with photosynthesis, representation by multiple alleles, co-expression with other photosynthesis genes, and overlap with findings from other photosynthesis mutant libraries. These HC genes are composed of three different statuses: 55 known genes, 29 inferred genes, and 170 unknown genes (1). The known status signifies the plentiful amount of experimental data to support the annotations of those photosynthesis genes. The inferred status signifies the existence of annotations that indicate a strong association with photosynthesis. The unknown status signifies relatively low annotations of the genes. However, phenotypic data suggests that these unknown genes are involved in photosynthesis.

1.3 Current Approach and Project Goal

Others in the Joint Genomics Institute have been working to incorporate a multi-omics, data-driven approach using transcriptomics, proteomics, and metabolomics to predict

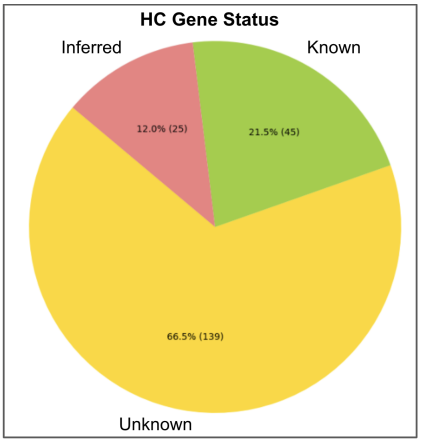


Figure 1: The status of these HC genes includes 55 known, 29 inferred, and 170 unknown genes. This chart does not show exact numbers because structure prediction is still pending for some genes.

potential functions of HC photosynthesis genes. Other types of data have also been integrated to gain a more holistic view of all the information at our disposal. This paper will discuss the utilization of *C.reinhardtii* pairwise graph networks and other structural similarity data to annotate potential functions of the HC photosynthesis proteins.

2 Methods & Materials

2.1 *C.Reinhardtii* Pairwise Graph Network

Pairwise similarity graph networks are a helpful tool for researchers to analyze genes that have similar expression patterns, sequences, or structural features. These networks help identify protein clusters of genes that are related. Utilizing this information can reveal new attributes of unknown genes, providing clues about their functional roles based on relationships with known genes.

To generate a pairwise similarity network of *C.reinhardtii* genes including the HC genes, we used an open-source software platform called Cytoscape. This platform generates a visualization of interaction networks, based on structural models, that integrates gene annotations, expression profiles, and other protein attribute data (pathway, database, function, etc). The graph network is set up with nodes, representing each protein, and edges, representing the metrics used to connect the proteins (Ono, 2002). In our case, the metric used are e-values produced by a software called FoldSeek, a structure alignment tool (Van et al. 2023). The lower the e-value between two proteins, the greater the similarity between them. In our *C.reinhardtii* graph network, the e-value threshold is $10e-10$. This creates protein clusters of *C.reinhardtii* genes that are similar in structure. Looking at the annotations of structurally similar proteins in the same clusters as the HC proteins can reveal patterns and relationships that could help in identifying protein

function.

The main strategy used is to color overlay different protein attributes onto the graph network. Performing this graph network analysis using the color overlay of other protein attributes can give us more information about the annotations of structurally similar proteins in the same clusters as the HC proteins. Integrating this information with more external structural similarity data could enable the inference of potential unknown functions of the HC proteins.

2.2 ProteinCartography Pipeline

Integrating additional structural similarity data towards predicting the unknown functions of HC genes is crucial to gaining a more holistic understanding of the HC genes. The *C.reinhardtii* graph network only enables us to observe proteins that are structurally similar to the HC proteins within the *C.reinhardtii* genome.

ProteinCartography, a pipeline published by Arcadia Science, is a way to compare the HC proteins to proteins of structural similarity from other types of organisms (2). The pipeline starts with the pdb and fasta files of the proteins of interest: the HC proteins. The pipeline conducts a BLAST sequence search against the NCBI non-redundant database, as well as a Foldseek search across the AlphaFold/UniProt50 v4, AlphaFold/Swiss-Prot v4, and AlphaFold/Proteome v4 databases. These searches identify proteins based on both structural and sequence similarities to the input. With millions of protein entries, these extensive databases enable comparisons on a very large scale. Aggregating, filtering, and downloading the hits from these exhaustive protein searches enables the construction of an all-v-all similarity matrix that compares all the proteins to each other, seeking the best matches using Foldseek search. From these matches, a TM-score, or similarity score, is calculated. The score ranges from zero to one with a value of one indicating that two structures are identical and values closer to zero indicating less structural similarity. Using these scores, an all-v-all similarity matrix is constructed.

This matrix is used to cluster the proteins based on similarity through the use of two different clustering algorithms: Foldseek's TM-align greedy set clustering and the Leiden algorithm (default for visualizations). From the protein clusters, a cross-cluster similarity matrix is created which is a heatmap representing the average TM-score of all structures in each cluster versus that of all the other clusters. With metadata from UniProt and clustering data, the pipeline performs a semantic analysis for each protein cluster. The semantic analysis consists of determining the most common annotations per cluster and creating individual bar charts for them while the most commonly used words per cluster are shown as word clouds.

After all of these steps, the pipeline uses the constructed matrices and aggregated

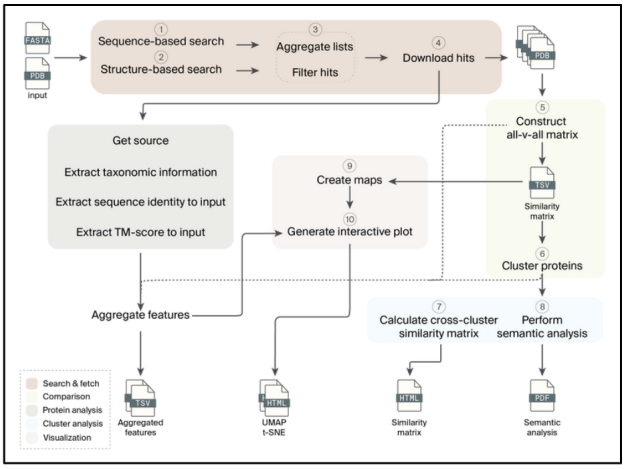


Figure 2: Flow chart of ProteinCartography pipeline in search-mode. Photo Credit: arcadiascience.com

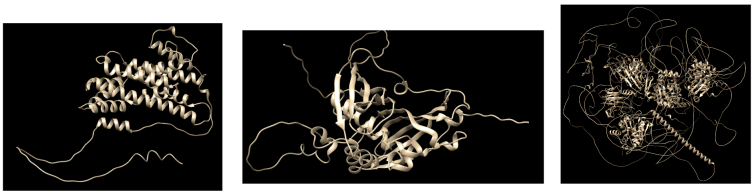
data to calculate the principal component analysis (PCA) used to perform dimension reduction through t-distributed stochastic neighbor embedding (t-SNE) and a uniform manifold approximation projection (UMAP). Integrating the use of the Plotly Python package with the 2-dimensional maps from t-SNE and UMAP, results in interactive, navigable maps with color overlays associated with the protein metadata including length, broad taxon, TM-score, source, average pLDDT, and more (Avasthi, 2023).

To understand the feasibility of this pipeline for the HC genes, the pipeline was first tested on a small set of three HC genes: Cre02.g084350_4532 and Cre06.g308100_4532, two genes of known status, and Cre10.g433900_4532, a gene of unknown status (3). After inputting these genes into the pipeline, interpretable interactive plots and semantic analyses were generated. These results introduced information about structurally similar proteins in other organisms per HC gene. Once this pipeline is applied to the entire large set of HC genes, merging the annotations of structurally similar proteins within and outside of the *C.reinhardtii* genome can reveal the information relevant for predicting potential functions.

3 Results

3.1 Patterns in *C. reinhardtii* Graph Network

Using the C.reinhardtii pairwise graph network generated by Cytoscape, we applied a color overlay based on protein pathway information. In Figure 4, colored nodes represent proteins with pathway information, while black nodes indicate proteins lacking such data. Examining the clusters within this pathway color overlay reveals two notable patterns.



Gene Name	Status	Description	Function
Cre02.g084350_4532	Known	Mn transport across thylakoid membrane	Photosynthesis
Cre06.g308100_4532	Known	Enoyl-CoA hydratase 2	Lipid FA metabolism
Cre10.g433900_4532	Unknown	NaN	Protease ubiquitin proteasome

Figure 3: Structures of test input genes from UCSF chimera and table containing information about the test input HC genes.

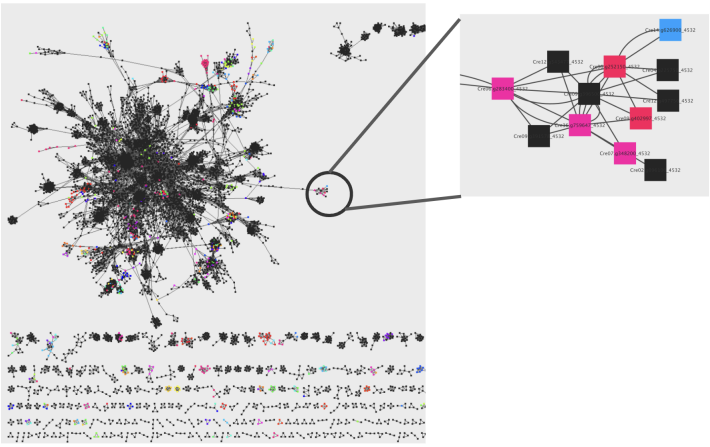
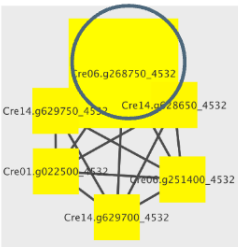


Figure 4: *C. reinhardtii* graph network, generated using Cytoscape, with color overlay of pathway information.

3.1.1 Proteins in Same Pathways in Single Cluster

The first pattern identified were that proteins belonging to the same pathway tend to cluster together. For instance, in Figure 5, the cluster containing the HC proteins of unknown function, Cre06.g268750_4532, also includes other proteins from the same pathway. Similarly, Figure 6 shows another cluster with an HC protein of unknown status, Cre09.g397956_4532. Although this HC protein lacks pathway information, most other genes in the cluster have known pathway annotations, potentially indicating the pathway association of Cre09.g397956_4532.



Name	Pathways
Cre06.g268750_4532 (Unknown)	• C4 photosynthetic carbon assimilation cycle, NAD-ME type • gluconeogenesis I
Cre14.g629750_4532 Cre14.g629700_4532 Cre14.g628650_4532 Cre01.g022500_4532 Cre06.g251400_4532	• C4 photosynthetic carbon assimilation cycle, NADP-ME type • gluconeogenesis I • C4 photosynthetic carbon assimilation cycle, PEPCK type

Figure 5: Cluster containing unknown HC protein (circled) in the same pathways as the other proteins in the cluster.

Name	Pathway
Cre09.g397956_4532 (Unknown)	NaN
Cre06.g278094_4532 (Unknown) Cre28.g802192_4532 Cre14.g632823_4532 Cre05.g800572_4532 ...	xylogalacturonan biosynthesis

Figure 6: Cluster containing unknown HC protein with no pathway information (circled) in the same cluster as other proteins in the same pathway.

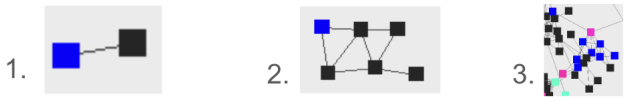
3.1.2 Proteins in Same Pathways Spread Across Different Clusters

The second pattern observed is the presence of proteins from the same pathways distributed across different clusters, as shown in Figure 7. This observation is consistent with the inherent complexity of biological pathways, where various proteins carry out specialized functions and exhibit diverse structural features. For instance, in the adenosine ribonucleotides biosynthesis pathway, enzymes facilitate the stepwise formation of adenosine monophosphate (AMP) from ribose-5-phosphate, a key building block for RNA synthesis and cellular energy transfer. Similarly, the superpathway of adenosine nucleotides biosynthesis I encompasses a broader network of reactions, integrating the production of AMP, ADP, and ATP, which are vital for cellular energy metabolism and signaling.

Further complexity arises in the superpathway of adenosine nucleotides biosynthesis II, which includes alternative routes and salvage pathways, allowing cells to efficiently recycle nucleotides and adapt to varying metabolic demands. The superpathway of purine nucleotides biosynthesis I, meanwhile, illustrates the interconnectedness of adenosine and guanosine nucleotide synthesis. Starting from a common precursor, inosine monophosphate (IMP), this pathway diverges to produce both AMP and guanosine monophosphate (GMP), highlighting the shared and distinct enzymatic steps that contribute to the overall nucleotide pool.

The distribution of proteins from these multifaceted pathways across different clusters in the network highlights the diverse roles these proteins play in cellular metabolism. This pattern, particularly observed in well-characterized pathways, supports the validity of the

graph network, reflecting the intricate and coordinated nature of metabolic processes at the molecular level.



Cluster	Gene Names	Pathway ID
1	Cre11.g467569_4532	• adenosine ribonucleotides de novo biosynthesis • superpathway of adenosine nucleotides de novo biosynthesis I • superpathway of adenosine nucleotides de novo biosynthesis II • superpathway of purine nucleotides de novo biosynthesis I
2	Cre17.g739550_4532	• adenosine ribonucleotides de novo biosynthesis • superpathway of adenosine nucleotides de novo biosynthesis I • superpathway of adenosine nucleotides de novo biosynthesis II • superpathway of purine nucleotides de novo biosynthesis I
3	Cre02.g081550_4532 Cre07.g325723_4532 Cre08.g377100_4532 Cre12.g494850_4532 Cre12.g519950_4532 Cre12.g557600_4532	• adenosine ribonucleotides de novo biosynthesis • superpathway of adenosine nucleotides de novo biosynthesis I • superpathway of adenosine nucleotides de novo biosynthesis II • superpathway of purine nucleotides de novo biosynthesis I

Figure 7: Proteins in the same pathways but spread across different clusters.

3.1.3 Importance of *C. reinhardtii* Graph Network Insights

These insights from the pathway color overlay provide valuable annotations and predictive information for HC genes, advancing our understanding of their potential roles and confirming the robustness of the network analysis. Further insights can be gained by applying color overlays of additional protein attributes onto the *C. reinhardtii* graph network, such as fold-change data and other predicted or inferred function information. Integrating these observations provides a comprehensive view of the relationships between well-characterized *C. reinhardtii* genes and the HC genes.

3.2 ProteinCartography Results

3.2.1 Inputting Each HC Gene into ProteinCartography Pipeline

While understanding the annotations of genes within *C. reinhardtii* is important, it is especially vital to observe the information of genes across different species that are structurally similar to the HC genes. Another focus of this research is examining the structural similarities between HC genes and those from other organisms using the ProteinCartography pipeline. Each HC gene with a predicted structure was individually input into the pipeline, which ran for approximately five days on a High Performance Computing cluster. The results yielded 210 interactive plots, each showcasing clusters of proteins from other organisms identified through BLAST and FoldSeek searches based on each HC protein.These plots can be colored based on various data types, including broad taxonomic classification, Leiden cluster, annotation score, sequence length, and TM-score. The goal is to understand the details of well-characterized proteins within the same clusters as the

HC genes. It is important to note that the utility and ease of analyzing these results depend significantly on the protein's structural characteristics. Intrinsically disordered proteins are much harder to analyze than structured proteins due to their lack of stability, less accurate structure predictions, and greater difficulty in alignment for structural comparisons.

3.2.2 ProteinCartography Results of DUF1995 HC Protein

One of the more structured HC proteins entered into the ProteinCartography pipeline is Cre03.g184550_4532 (8).

Input Protein	Status	Function	PFAM Description	Description
Cre03.g184550_4532	Known	Photosynthesis	DUF1995	Low Photosystem II Accumulation 3

Figure 8: Details about the input DUF1995 HC protein, Cre03.g184550_4532

This protein belongs to a group of Domain of Unknown Function 1995 (DUF1995) proteins, which are predominantly found in photosynthetic organisms and are currently under further analysis among the HC proteins. Notably, in Chlamydomonas reinhardtii, a DUF1995-containing protein known as LPA3 plays a crucial role in the assembly of photosystem II, a key complex in the photosynthetic process responsible for water splitting and oxygen evolution. The involvement of LPA3 in such a vital process suggests that other DUF1995 proteins may have similarly important roles, though much remains to be discovered.

The ProteinCartography results of Cre03.g184550_4532, shown in Figure 9, were selected for further analysis. After retrieving the interactive t-SNE plot for this protein, the most structurally similar protein was identified as A0A7J7QN35, a DUF1995 domain-containing protein from *Scenedesmus* sp. strain NREL 46B-D3, according to the AlphaFold database. *Scenedesmus* sp. is from the Chlorophyceae class in green algae. Although this strain has been sequenced at the Joint Genomics Institute, limited information is available about it. Nevertheless, this example highlights how sequencing diverse organisms can enhance large-scale analyses, such as the one conducted in this research.

3.2.3 UCSF Chimera Analysis

The structures of Cre03.g184550_4532 and A0A7J7QN35 were compared using UCSF Chimera, as shown in Figure 10. By manual inspection, the structures appeared very similar. A structural alignment comparison revealed a root mean square deviation (RMSD) of 0.7 angstroms between 278 pruned atom pairs. Additionally, the TM-score between

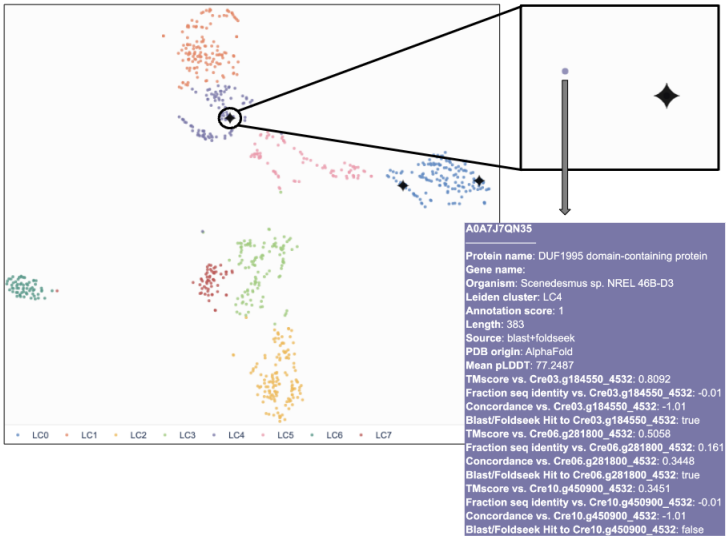


Figure 9: Interactive t-SNE plot, colored by leiden cluster, of protein hits based on input protein, Cre03.g184550_4532 (star). A0A7J7QN35 is the most structurally similar.

these two proteins was 0.81. These metrics statistically confirm the structural similarity between the two proteins, evidenced by the low RMSD and the high TM-score, which is close to one. However, the sequence similarity score between these two proteins is -0.1, indicating highly diverse sequences. When this type of analysis is applied to additional structured HC proteins, it not only deepens our understanding of the structural relationships among HC proteins but also highlights the importance of comparing gene structures across species to uncover potential functional insights.

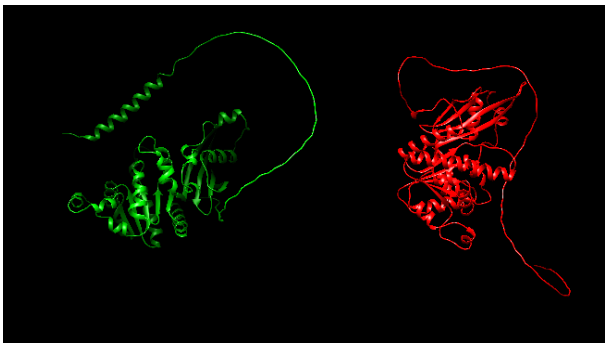


Figure 10: 3D structure of Cre03.g184550_4532 (right) and A0A7J7QN35 (left) from UCSF Chimera

4 Discussion and Conclusion

4.1 The Robustness of the *C. reinhardtii* Graph Network

The application of the *C. reinhardtii* pairwise graph network, enhanced with protein pathway color overlay, has elucidated interesting relationships among HC genes and structurally similar proteins within *C. reinhardtii*. The observed clustering patterns revealed that genes from the same pathway often are in the same cluster. Several underlying principles may explain this phenomenon. Biologically, genes involved in the same pathway are typically part of a coordinated system of interactions and functions. Proteins encoded by these genes often have similar roles, whether in enzymatic activity, regulatory functions, or interactions with other proteins. This functional similarity frequently extends to structural features, as proteins within the same pathway may share common domains crucial for their interactions and functions. Consequently, pathway-related genes may share similar structural characteristics, which can be leveraged to predict the functions of HC genes with unknown roles.

However, not all genes follow this pattern. Another observed pattern is that genes from the same pathways appear in different clusters. This underscores the complexity of biological pathways and indicates that predictions based on a single pattern are insufficient without additional information. Biological pathways are nuanced, typically involving a series of complex biochemical reactions and interactions among various proteins. Each protein in a pathway may have a unique role requiring different structural features to perform specific functions. Additionally, many proteins are composed of multiple domains, each performing distinct functions or interacting with different partners. These proteins can participate in multiple pathways, adding to the complexity of their roles.

Observing both these patterns in the *C. reinhardtii* graph network validates its accuracy and robustness. The presence of both patterns underscores the network's ability to capture the complexity and diversity of biological pathways more accurately than if only one pattern were observed. Moreover, this dual-pattern observation indicates that protein annotation is not straightforward; comprehensive and multi-source data are essential to make accurate predictions while minimizing biases.

4.2 Cre03.g184550_4532 and A0A7J7QN35 Comparison

Another type of data acquired is the cross-species structural similarity data from the ProteinCartography pipeline. Notably, the results from the well-structured DUF1995 HC gene Cre03.g184550_4532 were intriguing. The DUF1995 domain-containing protein, A0A7J7QN35, from the source organism, *Scenedesmus* sp., was found to be structurally similar to this input HC protein due to a low RMSD and TM-score close to one. This

similarity suggests that this protein could be instrumental in annotating the HC gene or providing deeper insights into the uncharacterized HC gene, thereby enabling more accurate function prediction in the future. Additionally, analyzing the taxon data of structurally similar proteins allows us to assess important phylogenomic patterns, which can provide broader evolutionary context and insights into gene function. However, further research on the *Scenedesmus* sp. protein is necessary to determine its feasibility for use in the final stages of analysis before establishing potential protein annotations.

Furthermore, A0A7J7QN35 belongs to the Chlorophyceae class in green algae, indicating that the DUF1995 motif is also present in this group. This is consistent with the fact that DUF1995 motifs are found in multiple photosynthetic organisms, including green algae and plants. For example, the most characterized DUF1995 protein is found in *Arabidopsis thaliana*, a model organism for plants. Results from the ProteinCartography pipeline consistently showed that proteins most structurally similar to the HC genes were classified as Viridiplantae, also known as "green plants." The defining characteristic of this group is the presence of chlorophylls a and b, essential for photosynthesis. This alignment with known biological classifications confirms the accuracy of the ProteinCartography pipeline in characterizing unknown HC proteins.

Additionally, an intriguing observation is that although A0A7J7QN35 and Cre03.g184550_4532 have similar structures, the sequence similarity score of -0.1 suggest that their sequences are markedly different. This sequence divergence can be attributed to several factors. One possibility is convergent evolution, where different sequences independently evolve to form similar structures because those structures confer functional advantages. Moreover, structural features are often highly conserved due to their necessity for protein stability and function, even amidst sequence variation. The folding of proteins is largely determined by the physical and chemical properties of amino acids rather than specific sequences, leading to similar folding patterns from different sequences. Structural similarity is a strong determinant of functional similarity, underscoring the utility of ProteinCartography in providing valuable insights into functional relationships among proteins.

4.3 Future Directions

Future actions will focus on comparing the ProteinCartography results with those obtained from BLAST and potentially other well-established structural comparison platforms. By running the HC genes through these platforms, we aim to determine if comparative structural analysis provides additional insights beyond what is achievable through sequence similarity alone, and to assess the extent of any additional information gained. To visualize multiple HC genes on a single interactive plot, we plan to cluster the HC genes based on structural similarity and then input them into the ProteinCartography pipeline in batches. Additionally, we will perform a detailed analysis of ProteinCartog-

raphy results from well-structured HC proteins using UCSF Chimera, collecting data on characterized proteins from other organisms that are structurally similar to the HC proteins. By integrating this information with annotations from structurally similar proteins within the *C. reinhardtii* graph network, which are color overlaid with various protein attributes, we aim to gather sufficient data to predict potential functions of the HC proteins.

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